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FPGA Design Using VHDL

Module 6B: Arithmetic Packages in VHDL

Which Arithmetic Package to Use?

- `numeric_std` vs `std_logic_arith`
- Approved IEEE package: `numeric_std`
- Old Defacto package: `std_logic_arith`
    - Written by Synopsys before there was a standard
- `numeric_std` guaranteed to be portable among synthesizers and simulators
- `std_logic_arith` various implementations between synthesis tools
- Recommendation : use approved IEEE package
    - `numeric_std` is preferred over `std_logic_arith`
- **Never use both together!**

Numeric Operation Packages

- `numeric_std`: IEEE standard
  - Defines types signed and unsigned
  - Defines arithmetic, comparison, and logic operators
- `std_logic_arith`: Synopsis
  - Defines types signed and unsigned
  - Defines arithmetic and comparison operators
- `std_logic_unsigned`: Synopsis
  - Defines arithmetic and comparison operators for `std_logic_vector`
- Recommendation: Use `numeric_std` package
- NEVER use both `numeric_std` and `std_logic_arith`
  - Compiler may get confused

Examples of Different Packages

```
library IEEE;
use ieee.std_logic_1164.all;
use ieee.numeric_std.all;

entity packages_example1 is
port (clk, reset : in std_logic;
        out_slv : out std_logic_vector(7 downto
0));
end packages_example1;

architecture arch of packages_example1 is

signal cntr : unsigned(7 downto 0);

begin

   process(clk, reset)
   begin
      if(reset = '1') then
         cntr <= (others => '0');
      elsif(rising_edge(clk)) then
         cntr <= cntr + 1;
      end if;
   end process;

   out_slv <= std_logic_vector(cntr);

end arch;
```

```
library IEEE;
use ieee.std_logic_1164.all;
use ieee.std_logic_arith.all;
use ieee.std_logic_unsigned.all;

entity packages_example2 is
port (clk, reset : in std_logic;
      out_slv : out std_logic_vector(7 downto 0));
end packages_example2;

architecture arch of packages_example2 is

signal cntr : std_logic_vector(7 downto 0);

begin

   process(clk, reset)
   begin
      if(reset = '1') then
         cntr <= (others => '0');
      elsif(rising_edge(clk)) then
         cntr <= cntr + 1;
      end if;
   end process;

   out_slv <= cntr;

end arch;
```